

Question	Marking Guidance	Mark	Comments
05.1	1. Use a grid OR Divide field/area into squares; 2. Method of obtaining random coordinates/numbers e.g. calculator/computer/random numbers table/generator; 3. Count number/frequency in a quadrat; 4. Large sample and calculate mean number (per quadrat); 5. Multiply mean number of plants per m² by area of field OR Divide area of field by area of quadrat x mean number of plants per quadrat OR Multiply mean number of plants per quadrat/section by number of quadrats/sections in field;	5 (5 x AO1)	1. Accept use of tape measures/map/area with coordinates. 1. Accept Belt transect. 2. If transect method used accept quadrats at regular intervals. 3. Ignore amount/abundance. 3, 4 and 5. Accept sections/squares for quadrats. 4 and 5. Accept 'average' for 'mean'. 4. Accept large sample and method of calculating mean. 4. Accept many/multiple for large sample but ignore several. 4. If a specific number is given it must be 10 or more. 5. Do not allow 'scale up' without further qualification.

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05.2	1. Interspecific (competition); 2. Less/no light (for potato plant) so less/no photosynthesis; 3. Less nitrates/nitrogen to produce amino acids/protein/DNA/RNA/ATP OR Less phosphate/phosphorus to produce DNA/RNA/phospholipids/RuBP/GP/triose phosphate/NADP/ATP;	3 (3 x AO2)	3. Accept any named organic nitrogen or phosphorus containing molecule.

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05.3	Correct answer of $6.69 / 6.7 = 2$ marks;; Incorrect answer but shows $669 / 67$ (ignore position of any decimal point or preceding / subsequent zeros or numbers following 669) = 1 mark OR Shows $0.012 = 1$ mark OR Shows $8.47 / 8.5 = 1$ mark OR Shows $8.69 = 1$ mark OR Shows 3.35×10^{-4} (in any correct mathematical form) = 1 mark;	2 (2 x AO2)	Ignore any numbers after 6.69 Ignore any numbers after 0.012 Ignore any numbers after 8.47 / 8.5 Accept answers which would round to 3.35×10^{-4} for 1 mark.